

Om Balar

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TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Swift, C, SQL

AI and ML: Claude, PyTorch, LLMs, RAG Architectures, Vertex AI / Google Cloud APIs

Infrastructure and Cloud: GCP, AWS, Docker, Kubernetes, Linux

Tools and Methodologies: AI-Assisted Coding (Claude Code, Cursor), CI/CD, Automated Testing, System Design

EDUCATION

Toronto Metropolitan University (Formerly Ryerson University)

Toronto, ON

Bachelor of Engineering, Computer Engineering (Software Engineering Specialization, With Co-op)

Sep 2022 - Apr 2027

- CGPA: **4.29/4.33**
- **Academic Honors:** Dean's List, Ontario Professional Engineers Foundation Scholarship (3 years in a row), Aditya Jha Undergraduate Diversity Award (Sole Recipient)
- **Relevant Coursework:** Operating Systems, Algorithms and Data Structures, Database Systems I, Intelligent Systems (Machine Learning)

WORK EXPERIENCE

Environment and Climate Change Canada

Toronto, ON

Software Developer

May 2025 - Present

- Heavily optimized database performance, implementing caching strategies that drastically reduced redundant queries and cut data processing time from **90 seconds down to 10 seconds**.
- Built a reliable **Python** and **SQL** data pipeline to process **roughly 1.5 million weather forecast files** with high data integrity.
- Acted with extreme ownership to **resolve end-user friction**, proactively building a custom formatting toolbar directly into an internal Markdown text editor to improve meteorologist workflows.
- Engineered automated CI/CD pipelines using **Docker** and **Cypress**, accelerating deployment velocity and cutting automated testing time from **12 hours to 7 hours**.

Reviewer.ly

Toronto, ON

Software Engineer

May 2024 - Aug 2024

- **Eliminated** the need for manual developer intervention by designing and shipping a complete self-service user profile system using **React**, **Spring Boot**, and **MongoDB**.
- Analyzed client-side **TypeScript** bottlenecks and optimized the application's full-stack performance by offloading heavy image compression workloads to **scalable concurrent backend servers**.
- Independently built an **automated SMTP email notification workflow** to alert engineers of platform issues, accelerating cross-team issue resolution.

DataAnnotation

Remote

AI Software Engineering Trainer

June 2026

- Evaluated AI coding assistants across **5 projects** by engineering targeted adversarial prompts to expose model failure modes, break logic, and identify hallucinations.
- Improved AI training data by reviewing AI-generated code and writing clear reports on system design mistakes.
- Built safe, isolated testing environments using **Docker** to run and check code written by AI models.

AI AND INFRASTRUCTURE PROJECTS

Omello | *Python, PyTorch, FastAPI, Swift* | [Live on the App Store](#)

Dec 2025 - Present

- Solo-developed a native macOS application featuring a **Swift** frontend and a **Python** backend, accelerating development velocity by natively integrating **Claude Code** into the engineering workflow.
- Built an advanced context-aware **RAG** architecture that seamlessly routes workloads between local models and cloud APIs.
- Fine-tuned a custom **PyTorch** neural network to operate securely on local hardware while strictly staying under a **500MB** RAM footprint constraint.
- Shrank the application payload size from **900MB down to 250MB** by aggressively optimizing the codebase and removing unused packages.

Distributed Job Search Platform | *React, Java, PostgreSQL, Kubernetes, Docker, GCP*

Jan 2025 - Apr 2025

- Built a reliable job search website using **React** and **Java** that lets users search through large amounts of data quickly.
- Kept the website online during heavy traffic by running **3 worker nodes** using **Docker** and **Kubernetes** on **Google Cloud Platform (GCP)**.